CE / CS / SE 3354 · SOFTWARE ENGINEERING · FALL 2026

Project Deliverable P1

Proposal and Requirements — Team Template

Weight 4% · Due Sunday 18 October 2026, 11:59 PM Central · Submit one PDF per team on Canvas

Field	Your entry
Project title	QR Code Restaurant Menu & Ordering System (QRMOS)
Team number and section	Team 01 · Section 003
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Repository URL	https://github.com/mrzetoon/team-01-QRRestaurantOrder
Document version	0.1
Date	18 October 2026

0 Revision History and Contribution Record

Both tables below are graded. The contribution record carries the individual 20% of this milestone and must be consistent with your repository history.

Version	Date	Author	Summary of change
0.1	2026-10-18	All members (see 0.1)	Initial baseline for P1

0.1 Individual contribution record

Team member	What this person produced for P1	Evidence in the repository
Mohmmad Abuzetoon	Sections 1.1, 1.2 and 4.1–4.4; set up the repository, issue board and branch rules; compiled and formatted the final PDF.	Commits by @mrzetoon: https://github.com/mrzetoon/team-01-QRRestaurantOrder/commits?author=mrzetoon; issues #1–#6; README.md, /docs/team-agreement.md, /docs/P1/member1_intro_scope_plan.md
Rafay Alvi	Section 3.1: wrote all 17 functional requirements (FR-01 to FR-17) with priorities, sources and acceptance criteria.	Commits by @MuhammadAlvi03 (functional requirements)
Len Isherwood	Sections 2.4, 3.2 and 3.3: constraints, the six quality requirements with thresholds and verification methods, and constraint requirements.	Commits by @ApePerson (quality & constraints)
Xuan Mai Tran	Section 2.1 and the 3.4 diagram: context diagram, use case diagram, and expanded use case UC-01 Place Order.	Commits by @keasark: github.com/mrzetoon/team-01-QRRestaurantOrder/commits?author=keasark; /docs/diagrams/context.png, /docs/diagrams/usecase.png
Eshitasri Punuganti	Section 3.5 and expanded use case UC-03 Process Incoming Orders; built the traceability matrix and found the missing table-setup requirement (FR-15).	Commits by @peshi24; /docs/P1/member5_UC03_traceability.md
Isaiah Bueno-Hernandez	Sections 1.3, 1.4, 2.2, 2.3, 2.5, 2.6 and 5: definitions, references, stakeholders, users, assumptions, feasibility and open questions.	Commits @buenoisaiah006-bit

1 Introduction

1.1 Purpose of this document and of the system

This document is the Phase 1 (P1) proposal and requirements baseline for the QR Code Restaurant Menu & Ordering System (QRMOS). It follows the section order of ISO/IEC/IEEE 29148 [R1] and is written to provide a detailed report to the restaurant manager to support their decision. It will grow into the full software requirements specification at P2.

QRMOS addresses the wait and the errors in ordering at a sit-down restaurant. A diner currently waits for a server to bring a menu, waits again to place the order, and depends on the server copying it correctly onto a ticket for the kitchen; at busy times this is where most delay and wrong orders come from. Printed menus also cannot show that an item has sold out or that a price has changed. QRMOS lets a diner scan a QR code on the table, read the current menu on their own phone, customize items, and send the order straight to the staff, tagged with the table number. Staff see orders arrive in real time and update their status; the manager keeps the menu current. The

problem belongs mainly to diners (waiting), servers and kitchen staff (transcription and running back and forth), and the owner (slow table turnover and menu reprinting costs).

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
A unique QR code per table that opens the menu for that table	Storing or processing card numbers ourselves — card entry happens only on the payment gateway's hosted page, which keeps card data out of our system [R5]
Digital menu by category with description, price, photo and availability	Integration with the restaurant's POS or accounting system — vendor APIs differ and need credentials we cannot get
Item customization (options, quantity, special note) and a cart	Takeout, delivery and table reservations — off-premises ordering is a different problem with different actors
Order submission tagged with the table number, plus order status for the diner	Customer accounts, loyalty points and reviews — not needed to order, and they add personal data we would have to protect
Live staff order queue with status updates (Received → Preparing → Ready → Served)	Native iOS/Android apps — a mobile web page needs no install, which is the point of scanning a code
Menu management, sold-out flags, table setup, and staff login with Manager/Staff roles	Multi-restaurant (franchise) management, inventory tracking and sales analytics
Paying the table's bill online through an external payment gateway (hosted checkout)	Splitting a bill between diners, tips and refunds — staff handle these at the till as today

Why the exclusions were excluded: each excluded item either brings legal or security obligations (storing card data, customer accounts) or depends on systems we do not control (POS), so including it would put the MUST requirements at risk inside one semester. None of them is needed for the core value: a diner can order from the table and the kitchen receives it correctly.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
QRMOS	QR Code Restaurant Menu & Ordering System — the system specified here.
QR code	Quick Response code: a 2-D barcode [R4] printed on a table card that a phone camera turns into a web link (URL).
QR-nn	**Not** a QR code: the identifier of a Quality Requirement in section 3.2 (template convention). Recorded here because the two meanings collide in this project.
Table session	The period from a diner opening a table's link until they close it or it expires; all orders in it carry that table number.
Order	One submission of a cart from one table, with its own order number. **Two meanings found:** staff also say "order" for the whole table's bill. We use the first meaning; see Q2 in section 5.
Order status	Where an order is in its life: Received, Preparing, Ready, Served, or Cancelled.
Modifier / option	A choice attached to a menu item, e.g. size, spice level, add extra cheese (+$1.00).
Sold out	An item that is shown on the menu but cannot be ordered right now. Different from **removed**, which means the item no longer appears at all.
Staff / Manager	Staff = server or kitchen employee who processes orders. Manager = owner or shift lead who can also edit the menu, tables and accounts.
Real time	In this document: visible on the other screen within 3 seconds without a page reload (QR-01).

Term or abbreviation	Meaning as used in this project
POS	Point of sale: the restaurant's existing till / card terminal. Outside QRMOS.
Payment gateway	An outside service (e.g. Stripe, Square) that shows its own card-entry page, charges the card and tells QRMOS approved or declined. A supporting actor in UC-07.
«include» / «extend»	Use case links: «include» = a step the base use case always uses (Log In, Browse Menu); «extend» = optional behaviour added only in some cases (Customize Item, Cancel Order).
FR, CON, UC, MoSCoW	Functional requirement; constraint requirement; use case; priority scale MUST / SHOULD / COULD / WON'T.

1.4 References

[R1] ISO/IEC/IEEE 29148:2018, Systems and software engineering — Life cycle processes — Requirements engineering.

[R2] CS 3354 Fall 2026 syllabus and P1 template; course textbook Chapters 4–6 (feasibility, requirements, use cases).

[R3] W3C, Web Content Accessibility Guidelines (WCAG) 2.1, Level AA.

[R4] ISO/IEC 18004:2015, QR Code bar code symbology specification.

[R5] PCI Security Standards Council, PCI DSS v4.0 (cited to justify keeping card entry on the gateway's page, CON-02).

[R6] Team brainstorm notes, 10 Oct 2026, /docs/elicitation/2026-10-10-brainstorm.md (restaurant interview planned for 23 Oct).

2 Overall Description

2.1 Product perspective and context

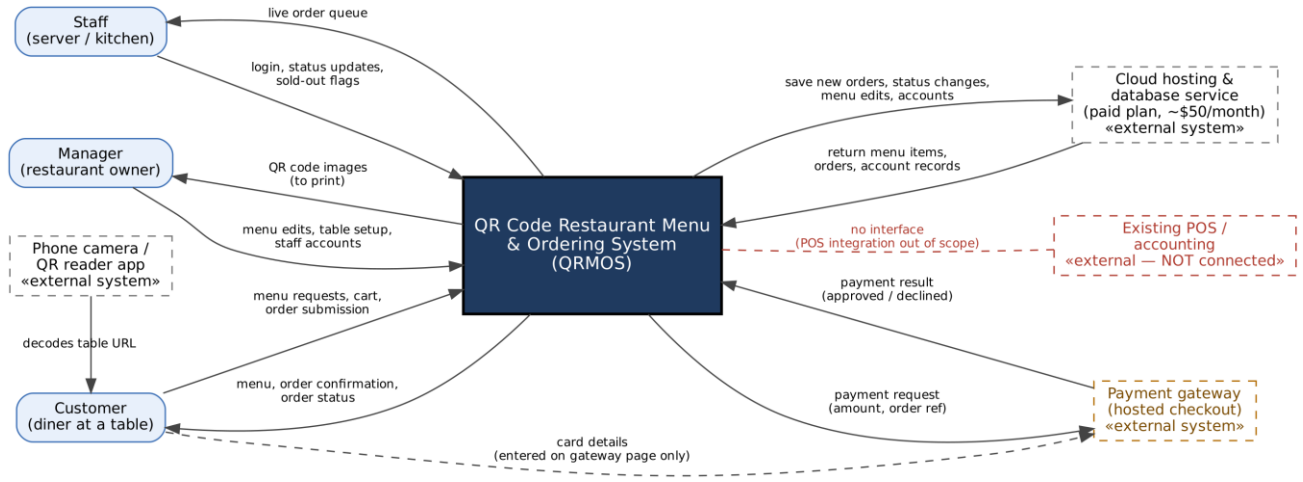


Figure 1. Context diagram. Everything inside the dark box is ours to build: the customer menu pages, the staff order queue, the manager pages, and the logic and data behind them. The boundary sits at the phone browser on one side and the hosting/database service on the other: we use the phone camera, the cloud host and the payment gateway but build none of them, so they are drawn as external systems. The customer types card details only on the gateway's own page; QRMOS sends the amount and receives approved or declined, so card data never enters our box. The existing POS is drawn on purpose with **no** interface, which makes that exclusion visible instead of silent.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
S1 Customer (diner)	Primary user; scans the code and orders	Order without waiting; see real prices and what is sold out; know the order was received	Assumed from team members' own dining experience; to be checked in a 10-person student survey by 25 Oct
S2 Server	User; delivers food, handles problems at the table	Fewer trips to take orders; orders clearly tagged by table	Assumed; interview with a partner restaurant planned 23 Oct (Q1–Q4)
S3 Kitchen staff	User; prepares food from the order queue	Legible orders with every option and note, in arrival order	Assumed; Isaiah Bueno-Hernandez has part-time restaurant experience — to be written up in /docs/elicitation
S4 Manager / owner	Pays for and operates the system; edits menu	Change menu and sold-out items instantly; faster table turnover; a build cost within budget and running cost under about $400/month	Assumed; to be confirmed at the 23 Oct interview
S5 CS 3354 instructor & TAs	**Never uses it**; assesses the project	Testable, traceable requirements; repository access; on-time milestones	Read from the syllabus and P1 template [R2]
S6 Information Technology (IT) team	Builds and maintains it this term	Scope that fits one semester (2 months) and a $8000-$75,000 budget (depends on case-by-case)	Agreed at team meeting weekly Saturday at 2:00pm, 2 hours, virtual online meeting on Microsoft Teams QR Code Ordering System for Restaurant Businesses
S7 Payment gateway provider	**Never uses the screens**; external service that processes card payments (supporting actor in UC-07)	Requests in its API format, a merchant account for the restaurant, its per-transaction fee	Assumed from providers' public developer documentation; which provider is open question Q1

2.3 User characteristics

Customers. Any adult or teenager in the restaurant, from tech-comfortable students to older diners who rarely scan QR codes. They use their own phone (any size, iOS or Android, any modern browser), often one-handed, in dim light and in a hurry to order before the rest of the table. They cannot be assumed to read English fluently, to have good eyesight, or to know restaurant terms. Some will use screen readers or large text, so the menu must work with both (QR-04). They will not create an account or install anything.

Staff (servers and kitchen). Know the menu and the restaurant's routine very well but have no training in this system and no time to read instructions during service. They will look at a shared tablet or screen from a distance, sometimes with wet or gloved hands, so the order queue needs large text and large buttons. Turnover is high, so a new staff member must be productive after a short demo.

Manager. Comfortable with a laptop and with editing a spreadsheet-like list, but not a programmer. Edits the menu occasionally, usually before opening, and sets up tables once. Needs to print QR codes on an ordinary office printer.

2.4 Constraints

C1 — Web application, no install. Customers must reach the menu through a phone web browser; no native app. Source: project definition agreed by the team and stated in our proposal.

C2 — Time. The system must be specified, designed, built and tested within the Fall 2026 term, by the P3 deadline. Source: syllabus [R2].

C3 — Budget. Total cost for building the system, its hardware and the first year of hosting must stay between $8,000 and $75,000. Our estimate is about $43,300 (see 2.6). Source: CS 3354 project instructions [R2].

C4 — No card data in our system. Payment must go through an external payment gateway's hosted page; QRMOS must never collect, transmit or store card numbers. Source: scope decision in 1.2 and PCI DSS [R5].

C5 — Instructor access. All source code and documents must live in a repository the instructor can open. Source: P1 template section 4.4 [R2].

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	Most diners carry a smartphone with a camera, a browser and mobile data or Wi-Fi.	Diners without a phone cannot order; the restaurant must keep paper menus and server ordering as a fallback, and adoption numbers fall.
A2	The restaurant has a reliable internet connection during service.	Orders are delayed or lost and the staff queue shows stale data; QR-01 cannot be met.
A3	The restaurant has at least one tablet or PC that stays open on the staff order screen.	No one sees incoming orders, so UC-03 has no actor at the moment orders arrive.
A4	(Dependency) A paid cloud hosting and database plan (about $50/month) stays available and priced as quoted for the first year.	We lose the demo environment and must migrate mid-project, threatening C2 and the P3 demo.
A5	(Dependency) The restaurant has, or can open, a merchant account with a payment gateway that offers a hosted checkout page.	Online payment (UC-07) cannot go live; diners pay the server as today, and FR-16/FR-17 slip to a later release (see Q1).
A6	A QR code on a table is only scanned by people sitting at that table.	Someone could photograph a code and send fake orders from outside; we would need staff confirmation of orders (see Q4).

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Feasible	Every piece is a common web feature: QR generation has free open-source libraries, live updates use WebSockets or 3-second polling, and menus/orders are plain database records. The IT team built a CRUD web app; a spike of QR → menu → live order is due 25 Oct in /spike-01.
Operational	Feasible, with one risk	It copies the existing workflow (order → kitchen → serve) and only moves where the order is typed. Staff need only a browser. Risk: older diners may not want to scan; A1 fallback keeps paper menus. Evidence is assumed until the 23 Oct interview.
Schedule	Feasible if scope holds	MUST items are held to about 50–60% of functional requirements (9 of 17), so the 7 SHOULD and 1 COULD give slack if work runs late; 6 use cases across about 7 weeks of build time; plan in 4.2 has dated owners. Risk R3 (scope creep) is managed by change control.
Economic / resource	Feasible	Estimated total about $43,300, within the $8,000–$75,000 budget (C3): • Development labour: 6 people × 10 weeks × 10 h = 600 h at an assumed $60/h = $36,000 • Cloud hosting and database, paid plan: $50/month × 12 = $600 • Domain name: $20/year • Payment gateway: no monthly fee assumed; the restaurant pays a per-transaction fee (about 3% per payment, to confirm in Q1) • Two staff tablets for the order screen: 2 × $350 = $700 • Printed, laminated QR table cards: 20 tables × $5 = $100 • Usability-test participant gift cards: 10 × $20 = $200 • Contingency 15% = $5,643 → total $43,263 Running cost after year 1: about $4,200/year (hosting $600, domain $20, maintenance about 10% of build $3,600). Rates are estimates to confirm (Q7).

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-01	The system shall generate a QR code for each registered table that encodes a URL unique to that table.	MUST	S4	Register tables 1–3, generate codes, scan each with a phone: each opens the menu showing its own table number, and no two codes decode to the same URL.
FR-02	The system shall display the menu, grouped by category, to anyone who opens a valid table URL without asking them to log in or install an app.	MUST	S1, C1	Open a table URL in a fresh private browser on iOS Safari and Android Chrome: the menu appears and no login or install prompt is shown.
FR-03	The system shall show each menu item's name, description, price and availability.	MUST	S1, S4	Load a 20-item test menu: every item on screen matches the database record in all four fields.
FR-04	The system shall prevent a customer from adding a sold-out item to the cart.	SHOULD	S1, S3	Mark one item sold out: its Add button is disabled, and a direct add request to the server is rejected.
FR-05	The system shall let a customer choose an item's defined options, quantity and a special note of up to 200 characters before adding it to the cart.	MUST	S1, S3	Add a burger with size Large, quantity 2 and a 200-character note: the cart shows all three; a 201st character cannot be typed.
FR-06	The system shall let a customer change the quantity of, or remove, any cart item before submission, recalculating the total after each change.	SHOULD	S1	Change 2 → 3 and remove one item: the displayed total equals the hand-calculated total after each step.
FR-07	The system shall submit the cart as one order tagged with the table number of the URL it was opened from.	MUST	S2, S3	Submit from table 5's URL: the order in the database and on the staff screen shows table 5.
FR-08	The system shall show the customer a confirmation with an order number after a successful submission.	SHOULD	S1	After submitting, the confirmation number on the phone matches the order number on the staff screen.
FR-09	The system shall add each new order to the staff order queue, in submission-time order, without the staff member reloading the page.	MUST	S2, S3	With the staff screen open, submit 3 orders from 3 phones: all 3 appear in the right order with no reload (timing tested in QR-01).
FR-10	The system shall let a staff member move an order's status from Received to Preparing to Ready to Served, or to Cancelled.	MUST	S2, S3	Step one order through every status and cancel another: each change is saved with a timestamp; skipping or reversing a status is refused.
FR-11	The system shall show the customer the current status of each order submitted from their device in the current table session.	SHOULD	S1	Change an order's status on the staff screen: the customer's page shows the new status within 3 seconds without a reload.
FR-12	The system shall let a manager add, edit and remove menu items, including name, description, price, category, options and photo.	MUST	S4	Add, edit and delete one item: each change appears on a freshly loaded customer menu.
FR-13	The system shall let a staff member or manager mark a menu item sold out or available again.	SHOULD	S2, S3, S4	Mark an item sold out on the staff screen: a freshly loaded customer menu shows it as sold out; mark it available and it can be added again.
FR-14	The system shall require a staff or manager account to log in with a username and password before any staff or manager page is shown.	MUST	S4, S5	Open every staff/manager page in a logged-out browser: each redirects to the login page.

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-15	The system shall let a manager add, rename and deactivate tables, and a deactivated table's URL shall stop accepting orders.	COULD	S4 (found via 3.5)	Deactivate table 4 and scan its code: the page says the table is not active and the Submit action is unavailable.
FR-16	The system shall let a customer pay all unpaid orders of their table session through the external payment gateway's hosted payment page.	SHOULD	S1, S4	Submit two orders from table 3 and choose Pay: the gateway page shows the combined total; a test card payment succeeds.
FR-17	The system shall mark the paid orders as Paid only after the payment gateway confirms the payment as approved.	SHOULD	S4	Pay with a test card the gateway declines: orders stay Unpaid and the customer is told; pay with an approved test card: orders show Paid on the staff screen.

3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	**Performance.** 95% of submitted orders shall appear on an open staff order screen within 3 seconds of submission, under 20 tables ordering at the same time.	MUST	S2, S3	Scripted test sends 100 orders from 20 simulated clients; compare submit and display timestamps; pass if ≥ 95 are ≤ 3 s.
QR-02	**Performance.** The customer menu shall show its first category of items within 3 seconds of opening a table URL in 95% of loads on a mid-range phone over a 4G connection.	MUST	S1	20 loads with Chrome DevTools "Fast 4G" throttling and a mid-range phone profile; record time to first item.
QR-03	**Usability.** At least 8 of 10 first-time users shall place an order of 2 items, one with an option chosen, in under 2 minutes without help.	SHOULD	S1	Moderated test with 10 students outside the team; timed from scan to confirmation screen; results in /docs/usability.
QR-04	**Accessibility.** The customer menu, cart and confirmation pages shall meet WCAG 2.1 Level AA [R3].	SHOULD	S1, R3	Zero serious/critical issues reported by axe DevTools, plus a full order completed using VoiceOver and TalkBack.
QR-05	**Security.** 100% of staff and manager server endpoints shall reject requests without a valid session, and manager-only endpoints shall reject Staff-role sessions.	MUST	S4	Automated test calls every endpoint logged-out, as Staff and as Manager; pass only if every expected 401/403 is returned.
QR-06	**Availability.** The system shall be reachable at least 99% of the time during a 7-day trial from 11:00 to 22:00 daily.	COULD	S4	An uptime monitor checks the menu URL every 5 minutes; uptime % computed from its log.

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The customer interface shall run in the phone's web browser without installing any app.	C1	It is the defining idea of the product; a diner will not install an app for one meal.
CON-02	The system shall not collect, transmit or store card numbers; card entry shall happen only on the payment gateway's hosted page.	C4, R5	Handling card data ourselves triggers full PCI DSS compliance the team cannot meet.
CON-03	The total cost of building the system, its hardware and the first year of hosting shall not exceed $75,000.	C3, R2	It is the budget ceiling set for the project; anything above it cannot be funded.
CON-04	All source code and documents shall be kept in a repository the instructor can access.	C5, R2	Required by the course for grading and contribution evidence.

3.4 Initial use cases

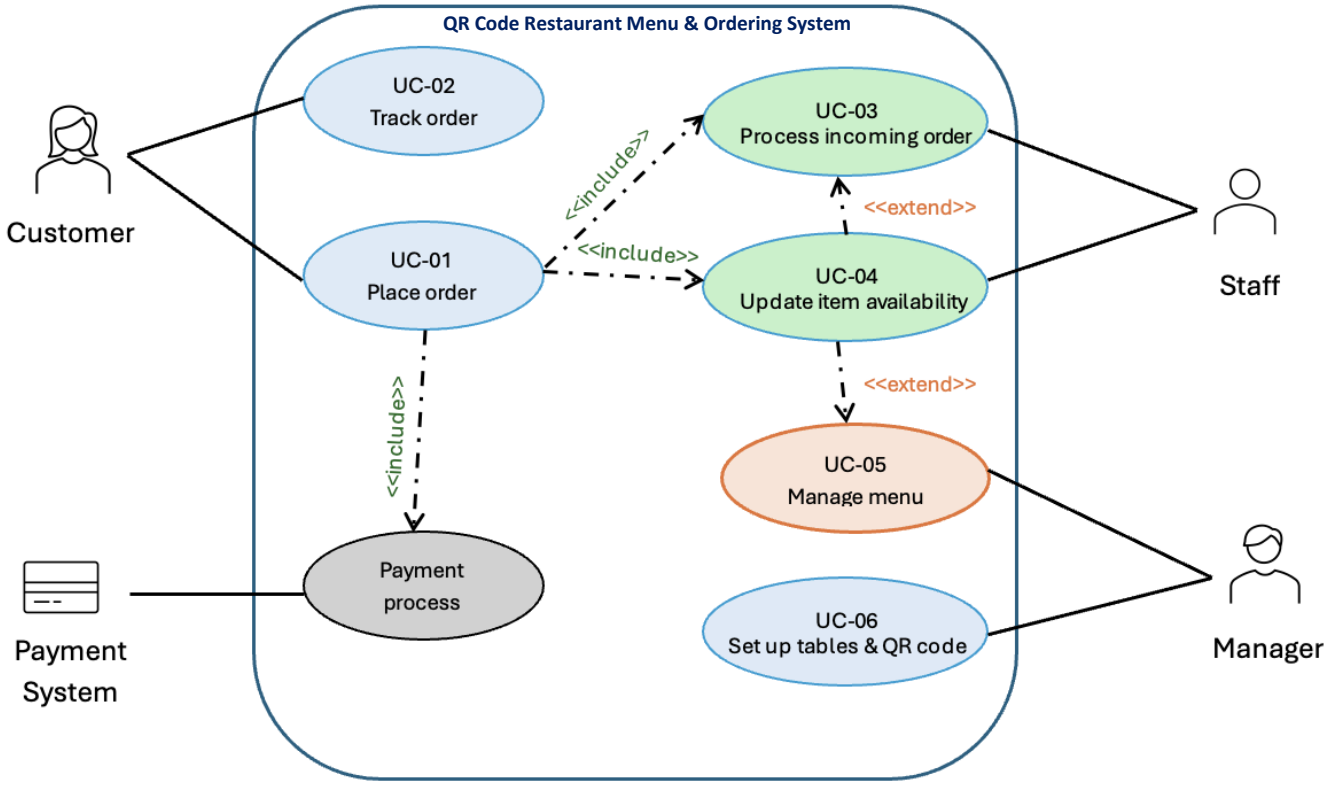


Figure 2. Use case diagram. Six actor-goal use cases. Manager is a specialized Staff actor, so a manager can also process orders and mark items sold out. Logging in is a precondition of UC-03 to UC-06, not a use case of its own, because nobody comes to the system just to log in. "Payment process" is NOT a use case, but it is a STEP of UC-01 and we won't build it. ***Note**: Primary actors include customer, staff (cashier, kitchen chef), and manager. Supporting actor includes the external system called "payment system" which we don't build but interacts closely to other use cases.*

Expanded use case 1 of 2 — UC-01 Place Order

Field	Content
Identifier and name	UC-01 Place Order
Primary actor	Customer
Supporting actors	Staff (receives the order through UC-03)
Preconditions — what must be true before it starts	1. The table is active and its QR code is on the table (UC-06). 2. The menu has at least one available item. 3. The customer's phone has a camera and internet access (A1).
Trigger — what starts it	The customer scans the QR code on their table.
Main success scenario — numbered steps	1. Customer scans the table's QR code. 2. System opens the menu for that table, showing the table number and the categories (Browse Menu «include»). 3. Customer opens a category and selects an item. 4. System shows the item's description, price and options.

Field	Content
	5. Customer chooses quantity and adds the item to the cart; if the item has options, Customize Item «extend» lets them pick options and type a note. 6. Steps 3–5 repeat until the customer has everything they want. 7. Customer opens the cart; System shows each item, its options, quantity and the total. 8. Customer submits the order. 9. System saves the order with the table number and status Received, and places it in the staff queue. 10. System shows a confirmation with the order number. The use case ends.
Alternative and exception flows	**2a. Table code not active:** System shows "This table is not taking orders — please ask your server"; no ordering is possible. Ends. **3a. Item is sold out:** System shows it greyed out and disables Add (FR-04); customer picks another item. **5a. Note longer than 200 characters:** System stops accepting characters at 200. **7a. Customer edits the cart:** customer changes a quantity or removes an item; System recalculates the total (FR-06). If the cart becomes empty, Submit is disabled. **8a. An item sold out after it was added:** System refuses that line, tells the customer which item, and keeps the rest of the cart for resubmission. **9a. Connection lost during submit:** System shows "Order not sent — try again" and keeps the cart. A retry must not create a duplicate order. ***a. Customer leaves without submitting:** the cart is discarded when the session expires; no order exists.
Postconditions — what is true when it succeeds	An order exists with a unique number, table number, items, options, notes, timestamp and status Received; it is visible in the staff queue; the customer holds its number.
Requirements realised	FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-15 (2a); QR-01, QR-02, QR-03, QR-04

Expanded use case 2 of 2 — UC-03 Process Incoming Orders

Field	Content
Identifier and name	UC-03 Process Incoming Orders
Primary actor	Staff member (server or kitchen); Manager may also act
Supporting actors	Customer (sees status changes through UC-02)
Preconditions — what must be true before it starts	1. The staff member is logged in (Log In «include», FR-14). 2. The staff order screen is open on the restaurant device (A3).
Trigger — what starts it	A new order arrives in the staff queue.
Main success scenario — numbered steps	1. System adds the new order to the queue in submission order, showing table number, items, options, notes and time (FR-09). 2. Staff member reads the order and marks it Preparing. 3. System saves the status with a timestamp and updates the customer's view (FR-10, FR-11). 4. When the food is ready, staff member marks the order Ready. 5. Server takes the food to the table and marks the order Served. 6. System moves the order from the active queue to the completed list. The use case ends.
Alternative and exception flows	**2a. Kitchen cannot make an item** (ingredient ran out) — Cancel Order «extend»: staff member marks the order Cancelled; System shows the cancellation to the customer; staff member marks the item sold out (UC-04) so it cannot be ordered again. Whether a single item can be cancelled instead is open (Q3). **2b. Order looks fake** (e.g. table is empty): staff member checks the table and cancels the order if no one is there (see A6, Q4). **3a. Staff tries to skip or reverse a status:** System refuses and keeps the current status (FR-10). **3b. Staff device loses its connection:** System shows an "Offline" banner; on reconnection it reloads the queue so no order submitted in the meantime is missed. **3c. Session expired:** System returns to the login page; after login the queue reopens unchanged.
Postconditions — what is true when it succeeds	The order's status is Served (or Cancelled) with a timestamp for every change; the customer's view shows the same status.
Requirements realised	FR-09, FR-10, FR-11, FR-13 (2a), FR-14; QR-01, QR-05

3.5 Requirements to use case traceability

Requirement ID	Use case(s) that realise it	If blank — why, and what you will do
FR-01	UC-06 (used by UC-01)	
FR-02	UC-01	
FR-03	UC-01	
FR-04	UC-01, UC-04	
FR-05	UC-01	
FR-06	UC-01	
FR-07	UC-01	
FR-08	UC-01	
FR-09	UC-03	
FR-10	UC-03	
FR-11	UC-02	Only requirement behind UC-02. If UC-02 fails, FR-11 does not show up / appear.
FR-12	UC-05	
FR-13	UC-04	
FR-14	Log In «include» of UC-03, UC-04, UC-05, UC-06	
FR-15	UC-06, UC-01 (alt 2a)	Added in this revision — see below.
FR-16	UC-07	
FR-17	UC-07	
QR-01 … QR-06	—	Non-functional by design: they constrain UC-01 and UC-03 rather than realise a use case, so they stay in 3.2.

What the matrix revealed, and what we changed because of it:

1. **UC-06 had nothing to trace to except FR-01.** FR-01 says "each registered table", but no requirement let anyone register a table. We added FR-15 (add, rename, deactivate tables) and the alternative flow 2a in UC-01.

2. **UC-02 rests on one SHOULD requirement (FR-11).** We kept it, but recorded that dropping FR-11 removes the whole use case.

3. **Role restriction first appeared as a functional requirement.** It reached no use case of its own, which showed it is a security quality, so it moved to QR-05.

4. **Ordering again from the same table** (a second round of drinks) has no requirement saying how it relates to the first order. Rather than invent an answer we raised it as Q2.

5. **Online payment had no use case.** Adding FR-16 and FR-17 needed a new goal, UC-07 Pay Bill, with the payment gateway as a supporting actor; CON-02 keeps card data on the gateway's page.

6. **Login was repeated in four use cases.** We made it one «include» (Log In) shared by UC-03 to UC-06, and split optional behaviour into «extend» parts (Customize Item, Cancel Order).

4 Project Plan

4.1 Team roles and working agreement

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Mohmmad Abuzetoon	Project manager and repository owner; editor of the final document	Discuss in the weekly meeting or on the team chat. If there is no agreement within 24 hours, majority vote; on a tie, the owner of the affected section decides. Every decision is logged as a GitHub issue labelled "decision".
Rafay Alvi	Functional requirements lead and change control	Same rule (agreed by all, /docs/team-agreement.md)
Len Isherwood	Quality and constraint requirements lead; runs the technical spike	Same rule (agreed by all, /docs/team-agreement.md)
Xuan Mai Tran	Modelling lead: diagrams and UC-01; leads P2 design models	Same rule (agreed by all, /docs/team-agreement.md)
Eshitasri Punuganti	Use cases and traceability; leads P2 architecture comparison	Same rule (agreed by all, /docs/team-agreement.md)
Isaiah Bueno-Hernandez	Elicitation and review lead: stakeholder interviews, feasibility, open questions, review checklist	Same rule (agreed by all, /docs/team-agreement.md)

4.2 Plan to P2

Date	What will be finished	Owner
18 Oct	P1 submitted on Canvas as one PDF	Mohmmad Abuzetoon
23 Oct	Interview with a restaurant manager and a server; notes in /docs/elicitation	Isaiah Bueno-Hernandez
25 Oct	Technical spike: QR scan → menu → order appears live on a second screen (/spike-01)	Len Isherwood
27 Oct	Q1–Q4 answered or re-dated; requirements revised under change control; revision history updated	Rafay Alvi
1 Nov	Two candidate architectures compared and one chosen (P2, Chapter 8)	Xuan Mai Tran
4 Nov	Class, sequence and state diagrams drafted for UC-01 and UC-03	Eshitasri Punuganti
6 Nov	Internal review of the P2 document against the rubric	Isaiah Bueno-Hernandez
8 Nov	P2 submitted	Mohmmad Abuzetoon

4.3 Risks

#	Risk	Likelihood	Impact	Mitigation — action and owner
R1	No restaurant agrees to an interview, so stakeholder needs stay assumptions.	Medium	High	Isaiah Bueno-Hernandez contacts three local or campus restaurants by 20 Oct; fallback is interviewing two students who have worked in restaurants by 25 Oct.
R2	Live order updates do not work on the chosen cloud host (connections time out or are blocked).	Medium	High	Len Isherwood builds the 25 Oct spike on the real free host; if WebSockets fail, switch to 3-second polling, which still meets QR-01.
R3	Scope creep — split bills, POS integration or loyalty get added.	Medium	Medium	Rafay Alvi runs change control: a new requirement needs an issue and a team vote, and is checked against the out-of-scope list in 1.2.

#	Risk	Likelihood	Impact	Mitigation — action and owner
R4	A member becomes unavailable or contribution becomes uneven.	Medium	High	Mohmmad Abuzetoon reviews the issue board every Monday, reassigns any task more than 3 days overdue, and reports persistent problems to the instructor as the syllabus asks.
R5	Labour runs over the estimate (more than 600 hours), pushing cost toward the budget ceiling.	Low	Medium	Xuan Mai Tran measures load time on the deployed site weekly from 1 Nov; if it fails, add a keep-alive ping or move to another free host.

4.4 Repository and tooling

Repository: https://github.com/mrzetoon/team-01-QRRestaurantOrder (owner @mrzetoon)

Branches: main is protected and only changes through pull requests; work happens on feature/<issue-number>-<short-name>. Each pull request needs one review from another member before merge.

Commits: start with the requirement or issue they touch, e.g. "FR-07: tag order with table number (#12)".

Issues: every task, requirement change and team decision is an issue, labelled requirement, doc, bug, spike or decision, with an assignee and due date.

Folders: /docs (P1–P3 documents, diagrams, elicitation notes, meeting minutes), /spike-01, /src (from P2).

4.5 Declaration of AI tool use

Tool	What it was used for	Who used it	How the output was checked
Claude (Anthropic)	Drafting a first version of the P1 content from our project description: requirement wording, use cases, diagrams and plan structure.	Team-01	Every requirement re-checked by the team against the testability test; assumptions and stakeholder sources rewritten to reflect what we actually did; references verified; placeholders replaced with our own data. Prompts and outputs saved in /docs/ai-log.

5 Open Questions

#	The question	Who can answer it	Needed by	What you assumed in the meantime
Q1	Which payment gateway does the restaurant already use or prefer, and who pays its per-transaction fee?	Restaurant owner (S4)	27 Oct	A gateway with a hosted checkout page; the restaurant pays about 3% per payment (A5, CON-02).
Q2	When a table orders a second time, is that a new order or an addition to the first? (Staff use "order" to mean the whole bill — see 1.3.)	Manager and a server (S2, S4)	27 Oct	Each submission is a separate order with its own number; UC-07 pays all unpaid orders of the table session together.
Q3	Can the kitchen cancel one item of an order, or only the whole order?	Kitchen staff (S3)	27 Oct	Whole order only (UC-03 alt 2a).
Q4	Must a server confirm each order before the kitchen sees it, to stop fake orders from a photographed QR code?	Restaurant manager (S4)	1 Nov	No confirmation step; staff can cancel suspicious orders (A6, UC-03 alt 2b).
Q5	Which languages must the menu support?	Restaurant manager (S4)	1 Nov	English only.
Q6	Must allergen information be shown in a fixed format?	Restaurant manager (S4)	1 Nov	Allergens are written in the item description text.

#	The question	Who can answer it	Needed by	What you assumed in the meantime
Q7	Will the restaurant fund about $43,300 to build and about $4,200 a year to run, and is $60/h a fair labour rate?	Restaurant owner (S4) and instructor (S5)	27 Oct	The estimate in 2.6 stands, within the $75,000 ceiling.

6 Baseline Declaration

We confirm that the requirements in section 3 represent our agreed baseline as of the submission date, and that every team member has read this document in full.

Team member	Signature or typed name	Date
Mohmmad Abuzetoon	Mohmmad Abuzetoon	18 Oct 2026
Rafay Alvi	Rafay Alvi	18 Oct 2026
Eshitasri Punuganti	Eshitasri Punuganti	18 Oct 2026
Xuan Mai Tran	Xuan Mai Tran	18 Oct 2026
Len Isherwood	Len Isherwood	18 Oct 2026
Isaiah Bueno-Hernandez	Isaiah Bueno-Hernandez	18 Oct 2026